WHEN THE TENSOR PRODUCT OF TWO GLUED DIMENSION-ONE LOCAL RINGS MISSES K THERE IS A COMMON DOMINATING DVR

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement sits in Situation situation-glue (the glueing setup for dimension-one rings: rings A, B with fraction fields mapping to K and R = A + B style notation).

Allowed standard prerequisites (editorial).

- Existence of a valuation ring dominating a local subring of a field. Algebra (valuation ring domination). Provides O dominating C_m.
- Krull-Akizuki. Algebra, Lemma algebra-lemma-krull-akizuki. Noetherianity of O.
- Noetherian valuation rings are DVRs. Algebra, Lemma algebra-lemma-valuation-ring-Noetherian-discrete. Final identification.

Source reference. Original reference 09MY: sources/stacks-chow/tags.txt, source lines 12196–12196. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00PG: sources/stacks-chow/tags.txt, source lines 930–930. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00II: sources/stacks-chow/tags.txt, source lines 680–680. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 09MY assume A is a Noetherian ring of dimension* 1*. The following are equivalent*

(1) *$A \otimes B \to K$ is not surjective,*
(2) *there exists a discrete valuation ring $\mathcal{O} \subset K$ containing both A and B.*

Proof. It is clear that (2) implies (1). On the other hand, if $A \otimes B \to K$ is not surjective, then the image $C \subset K$ is not a field hence C has a nonzero maximal ideal $\mathfrak{m}$. Choose a valuation ring $\mathcal{O} \subset K$ dominating $C_{\mathfrak{m}}$. By Algebra, Lemma 00PG applied to $A \subset \mathcal{O}$ the ring $\mathcal{O}$ is Noetherian. Hence $\mathcal{O}$ is a discrete valuation ring by Algebra, Lemma 00II. □

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